Setting up a Security Testbed in Windows Using Docker

Docker's versatility and cross-platform support make it essential for developers, DevOps engineers, and systems administrators. This tutorial will guide you through setting up Docker on your operating system and introduce a complete security testing environment using Docker. By the end of it, you'll have a functional test bed ready to explore vulnerabilities and security tools in a streamlined containerised environment.

Docker has revolutionised how developers and IT professionals deploy and manage applications, including security environments. According to a 2024 survey, over 70% of organisations now use containerisation, with Docker being the leading platform, powering millions of applications across different infrastructures. Docker's ability to create isolated, replicable, and scalable environments makes it ideal for setting up security test beds. Whether you are a cybersecurity enthusiast or a professional looking to test vulnerabilities, Docker allows you to deploy complex environments like Kali Linux, Damn Vulnerable Web Application (DVWA), and Metasploitable across Windows, Linux, and macOS.

Setting up a security test bed using Docker

After successfully installing Docker Desktop on your system, the next step is to build a security test bed using Docker Compose, a powerful orchestration tool that simplifies the management of multi-container Docker environments. Docker Compose is particularly beneficial when you need to run multiple containers that must interact with each other, such as in a security test bed where you may be working with different vulnerable applications and security tools. By defining your setup in a simple YAML configuration file (compose.yml), Docker Compose allows you to easily deploy and manage multiple containers with a single command, making the process efficient and repeatable.

In this security lab, we will deploy three essential security-focused containers.

Kali Linux: A Debian-based Linux distribution known for penetration testing and security auditing. It comes with hundreds of pre-installed tools covering areas like network scanning, password cracking, forensics, and reverse engineering, making it a must-have for security professionals and researchers.

Damn Vulnerable Web Application (DVWA): DVWA is a web application intentionally designed to be vulnerable, allowing users to test their skills in exploiting common web application vulnerabilities like SQL injection, cross-site scripting (XSS), file inclusion, and more. It's a safe environment to practice attacks and understand how to defend against them.